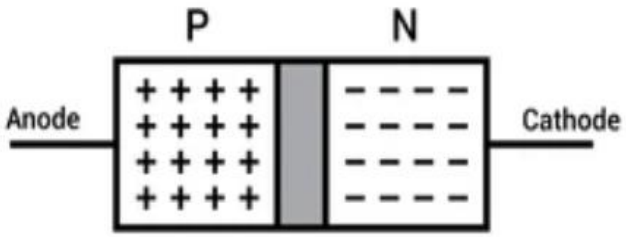
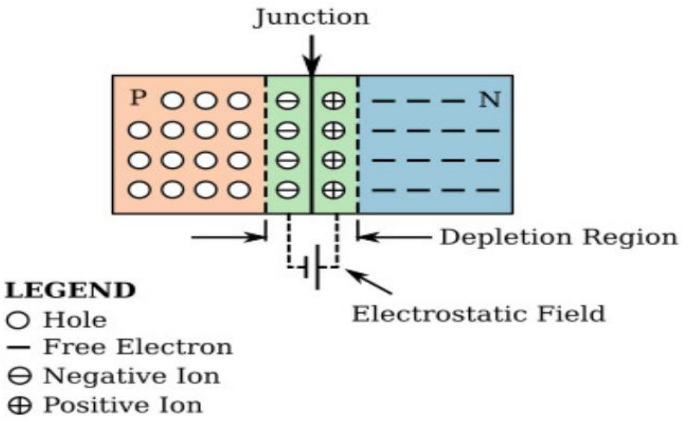
1. Explain the operation of **P-N junction diode** and draw **V-I characteristics**.

A PN junction is defined as an interface between two types of semiconductor materials (p-type and n-type semiconductors) inside a single crystal of semiconductor. The positive ("p") side contains an excess of holes, while the negative ("n") side contains an excess of electrons in the outer shells of atoms within the semiconductor.



At the instant of pn-junction formation, the free electrons near the junction in the n region begin to diffuse across the junction into the p region where they combine with holes near the junction. The result is that n region loses free electrons as they diffuse into the junction. This creates a layer of positive charges (pentavalent ions) near the junction. As the electrons move across the junction, the p-region loses holes as the electrons and holes combine. The result is that there is a layer of negative charges (trivalent ions) near the junction. These two layers of positive and negative charges form the **depletion region (or depletion layer).** The term depletion is due to the fact that near the junction, the region is depleted (i.e. emptied) of charge carries (free electrons and holes) due to diffusion across the junction. It may be noted that depletion layer is formed very quickly and is very thin compared to the n-region and the p-region.



Once PN junction is formed and depletion layer created, the diffusion of free electrons stops. In other words, the depletion region acts as a barrier to the further movement of free electrons across the junction. The positive and negative charges set up an electric field. There exists a potential difference across the depletion layer and is called **barrier potential**.

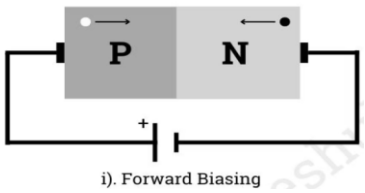
Operation of PN Junction Diode

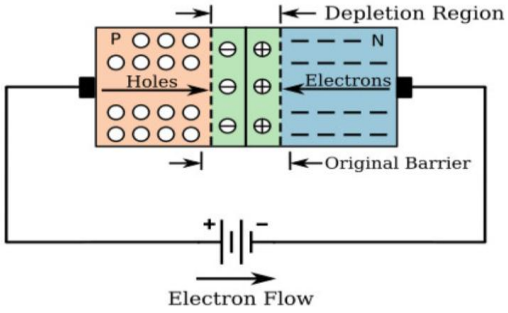
When an external dc source is applied across the PN Junction is called as biasing of PN Junction. By biasing the PN junction, the width of the depletion region can be controlled. The PN junction can be biased in two ways, such as

(i) Forward bias,
(ii) Reverse bias.

i) Forward bias

In the forward bias of a PN junction, positive terminal of the battery is connected to the p-side and the negative terminal to the n-side.





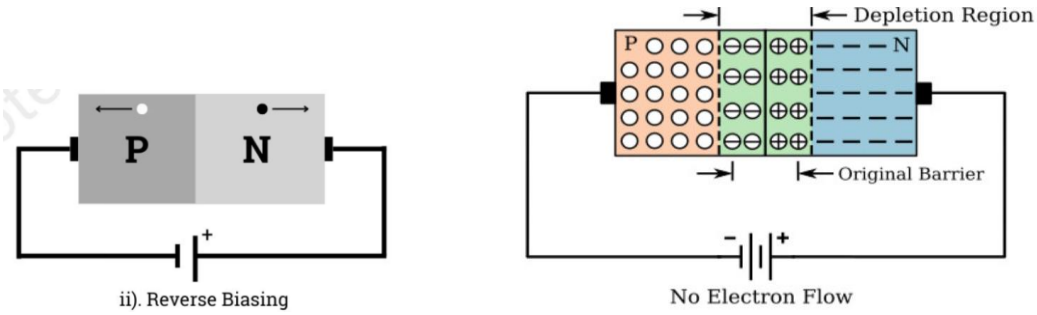
(i) The **free electrons** from the negative terminal of the battery continue to pour into the n-region while the free electrons in the n-region move towards the junction.
(ii) The electrons travel through the n-region as free-electrons i.e. **current in n-region is by free electrons.**
iii) When these electrons reach the junction, they combine with holes and become valence electrons.
(iv) The electrons travel through p-region as valence electrons i.e. **current in the p-region is by holes.**
(v) When these valence electrons reach the left end of crystal, they flow into the positive terminal of the battery.

It is concluded that in n-type region, current is carried by free electrons whereas in p-type region, it is carried by holes. However, in the external connecting wires, the current is carried by free electrons.

ii) Reverse bias

In reverse bias, connect negative terminal of the battery to p-type and positive terminal to n-type. The negative potential attracts the holes away from the edge of the junction barrier on the P side, while the positive potential attracts the electrons away from the edge of the barrier on the N

side. This action increases the barrier width because there are more negative ions on the P side of the junction, and more positive ions on the N side of the junction. Notice in the figure the width of the barrier has increased. This increase in the number of ions prevents current flow across the junction by majority carriers. However, in practice, a very small current (of the order of μA) flows in the circuit with reverse bias as shown in the reverse characteristics in Figure 1. This is called **reverse saturation current** (Is) and is due to the minority carriers.



V-I Characteristic Curve of PN junction Diode:

With forward bias to the pn junction i.e. p-type connected to positive terminal and n-type connected to negative terminal, the potential barrier is reduced. At some forward voltage (0.7 V for Si and 0.3 V for Ge), the potential barrier is altogether eliminated and current starts flowing in the circuit. From now onwards, the current increases with the increase in forward voltage. Thus, a rising curve OB is obtained with forward bias in Figure 1. The characteristic curve is drawn between Forward voltage (V_R) and Forward Current (I_F).

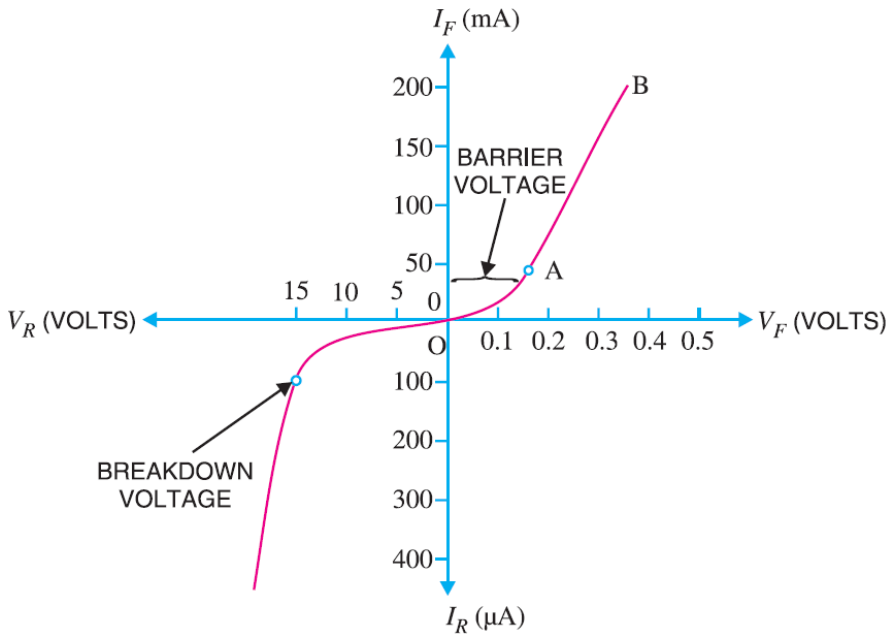


Figure 1. V-I Characteristic Curve of a PN Junction diode

If reverse voltage is increased continuously, the kinetic energy of electrons (minority carriers) may become high enough to knock out electrons from the semiconductor atoms. At this stage breakdown of the junction occurs, characterised by a sudden rise of reverse current and a sudden fall of the resistance of barrier region. This may destroy the junction permanently.

Diode Current Equation:

It describes the diode behaviour in both forward and reverse bias regions.

$$I_D = I_S \left(e^{qV_D/nKT} - 1\right),$$

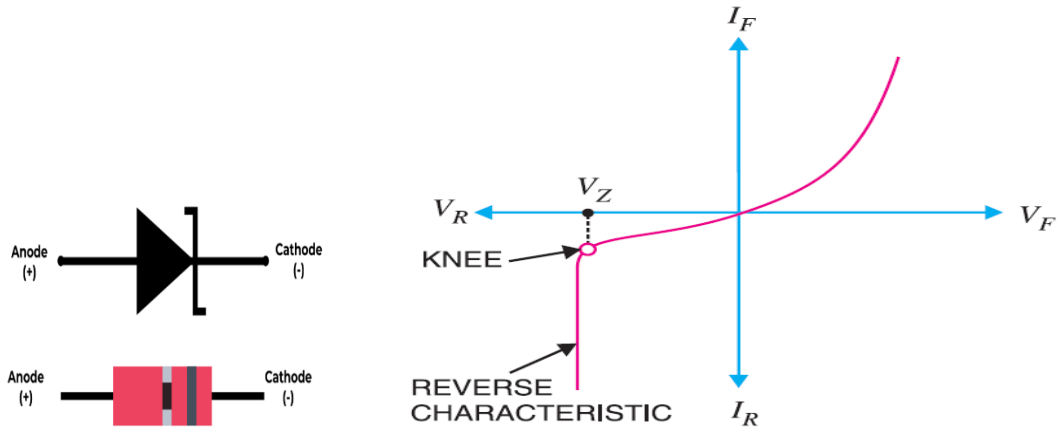
Where:

- I = current flowing through the diode
- IS = reverse saturation current (or dark saturation current), typically in the range of picoamperes (pA) for silicon diodes
- V = voltage applied across the diode
- n = ideality factor (also called emission coefficient), usually between 1 and 2
- VT = thermal voltage, given by $VT=qkT$
- k = Boltzmann's constant ($1.38\times10-23J/K$)
- T = absolute temperature in Kelvin
- q = elementary charge ($1.602\times10-19C$)

At room temperature (300 K), the thermal voltage VT is approximately **25.87 mV**.

2. Explain the operation of **zener diode** and draw **V-I characteristics**.

   A zener diode is a special type of diode that is designed to operate in the reverse breakdown region. An ordinary diode operated in this region will usually be destroyed due to excessive current. This is not the case for the zener diode. A zener diode is heavily doped to reduce the reverse breakdown voltage. This causes a very thin depletion layer. As a result, a zener diode has a sharp reverse breakdown voltage V_Z.



Operating Principle

The Zener diode when it works in reverse bias provides a large amount of current at a breakdown voltage point. The electrical breakdown can happen in two approaches such as

(i) Zener breakdown, and

(ii) Avalanche breakdown.

i) Zener Breakdown: The Zener breakdown occurs in the Zener diodes having V_z less than 6V or less. In Zener diode, the P and N regions are heavily doped and direct rupture of covalent bonds takes place due to the strong electric fields, at the PN junction. The electrons, thus free are available for conduction. The increase in free electrons contributes to large reverse current at a constant value of reverse bias and breakdown is said to have occurred due to **Zener effect**.

(ii) Avalanche Breakdown: The Avalanche breakdown takes place in Zener diodes having V_z more than 8V. In the reverse bias, the conduction is due to the minority carriers. As the reverse bias increases, these minority carriers tend to accelerate. The kinetic energy associated with them increases. These accelerated electrons rupture the covalent bonds and create new electron hole pairs. The new carriers upon acquiring sufficient energy from the field collide with other immobile ions to generate further electron hole pairs. This phenomenon is cumulative in nature, and the mechanism of carrier generation is known as **Avalanche multiplication**. A large amount of current flows at the same value of reverse bias due to the **Avalanche breakdown**.

V-I Characteristics of Zener Diode

The behavior of Zener diode is similar to that an ordinary diode in forward bias and the behavior in reverse bias is based on the breakdown phenomenon.

Forward bias of Zener diode: When the anode of the Zener diode is connected to the positive terminal of the de source and the cathode is connected to the negative terminal, then the Zener diode is said to be forward biased. The behavior of forward biased Zener diode is identical to the forward biased ordinary diode. The circuit of forward bias is shown in figure

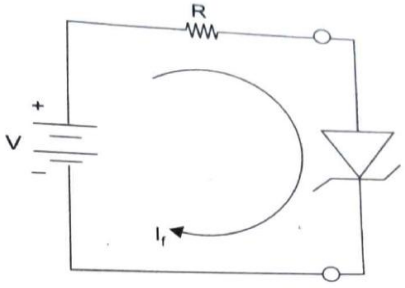


Figure (6.15): *Zener Diode in forward bias*

Reverse bias of Zener diode: When the cathode is connected to the positive terminal and anode is connected to the negative terminal of the de source, the Zener diode is said to be reverse biased. The circuit of reverse bias is shown in figure . In the reverse bias if the voltage applied is increased, the junction breaks down and reverse current increase sharply and is limited by the external resistance connected in series. The voltage at this point is called **breakdown voltage (V_Z)**. The breakdown voltage depends on the width of the depletion layer. This width of the depletion layer depends upon the doping level. There are two mechanisms by which the Zener diode can breakdown, such as (i) Zener breakdown and (ii) Avalanche breakdown.

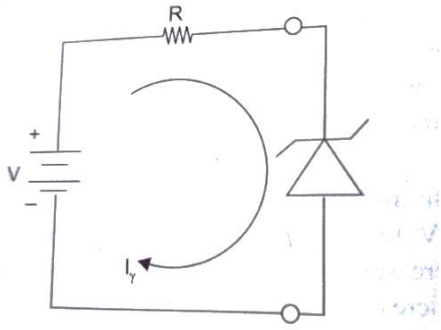
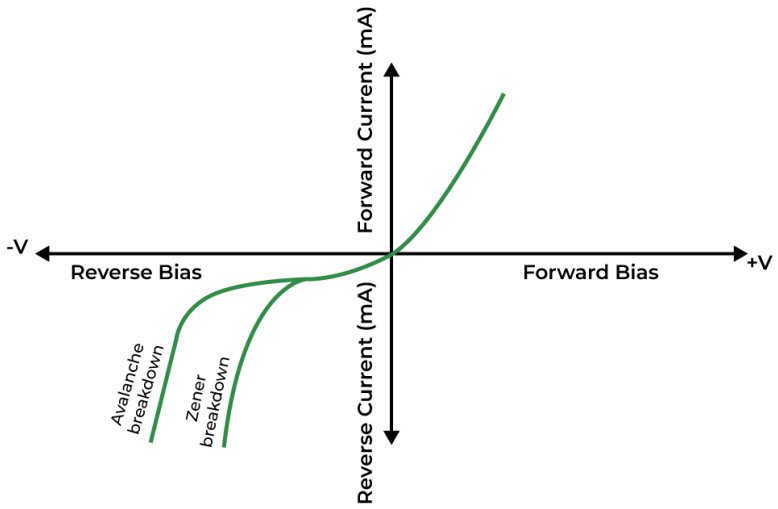


Figure (6.16): *Zener Diode in reverse bias*



3. Explain how **zener diode** acts as a **zener voltage regulator** with neat circuit diagram.

The zener diode operated in this region will have a relatively constant voltage across it, regardless of the value of current through the device. This permits the zener diode to be used as a voltage regulator. The circuit arrangement is shown in Figure (i). The Zener diode of zener voltage V_Z is reverse connected across the load RL across which constant output is desired. The series resistance R absorbs the output voltage fluctuations so as to maintain constant voltage across the load. It may be noted that the zener will maintain a constant voltage V_Z (= E_0)across the load so long as the input voltage does not fall below V_Z.

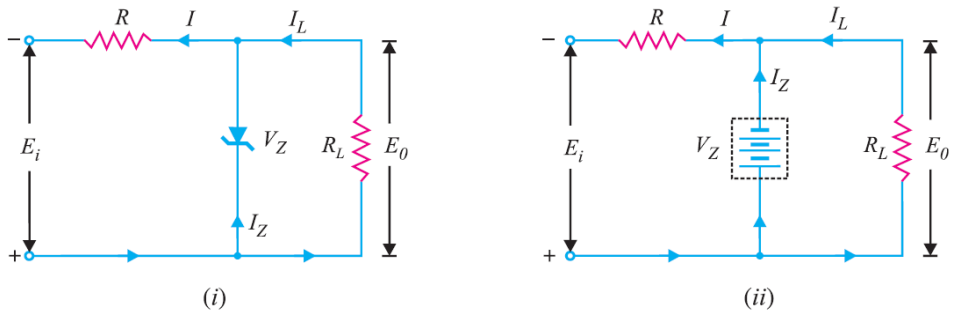


Figure (i). Zener Voltage regulator circuit arrangement in Reverse Bias.

When the circuit is properly designed, the load voltage E0 remains essentially constant (equal to V_Z) even though the input voltage Ei and load resistance RL may vary over a wide range.

(i) Suppose **the input voltage increases**. Since the zener is in the breakdown region, the Zener diode is equivalent to a battery V_Z. It is clear that output voltage remains constant at V_Z (= E_0). The excess voltage is dropped across the

series resistance R. This will cause an increase in the value of total current I. The zener will conduct the increase of current in I while the load current remains constant. Hence, output voltage E0 remains constant irrespective of the changes in the input voltage Ei.

- Voltage drop across $R = E_i - E_0$

(i) suppose that input voltage is constant but the load resistance R_L decreases. This will cause an increase in load current. The extra current cannot come from the source because drop in R (and hence source current I) will not change as the zener is within its regulating range. The additional load current will come from a decrease in zener current I_Z. Consequently, the output voltage stays at constant value.

- Current through R, $I = I_Z + I_L$.

4. Explain the operation of **NPN & PNP transistor**.

A transistor consists of two **pn** junctions formed by *sandwiching either p-type or n-type semiconductor between a pair of opposite types. Accordingly ; there are two types of transistors, namely;

(i) **n-p-n transistor (ii) p-n-p transistor**

An n-p-n transistor is composed of two n-type semiconductors separated by a thin section of ptype as shown in Fig. (i). However, a p-n-p transistor is formed by two p-sections separated by a thin section of n-type as shown in Fig. (ii).



(*i*) (*ii*)

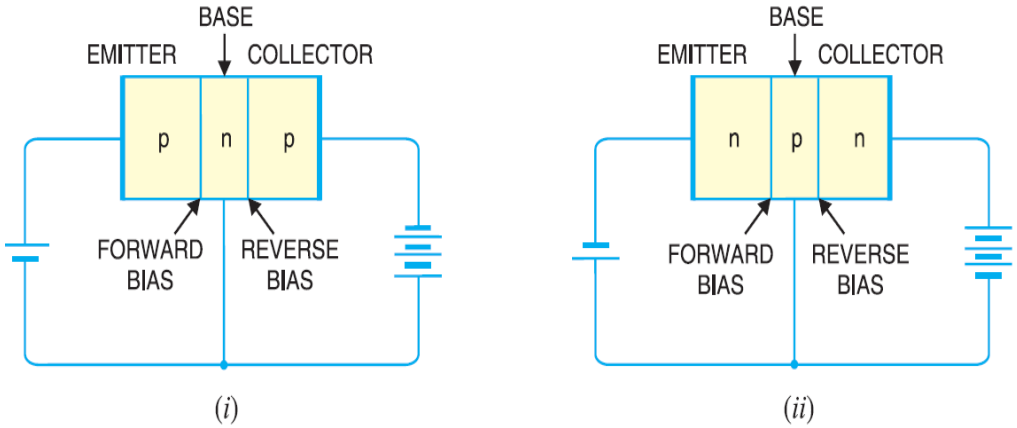
A transistor (*pnp* or *npn*) has three sections of doped semiconductors. The section on one side is the *emitter* and the section on the opposite side is the *collector*. The middle section is called the *base* and forms two junctions between the emitter and collector.

(i) **Emitter.** The section on one side that supplies charge carriers (electrons or holes) is called the *emitter*. *The emitter is always forward biased with respect to base* so that it can supply a large number of majority carriers. In Fig. *(i),* The emitter (*p*-type) of *pnp* transistor is forward biased and supplies hole charges to its junction

with the base. Similarly, in Fig. (*ii*), the emitter (*n*-type) of *npn* transistor has a forward bias and supplies free electrons to its junction with the base.

(ii) **Collector.** The section on the other side that collects the charges is called the *collector*. *The collector is always reverse biased*. Its function is to remove charges from its junction with the base. In Fig. (*i*), the collector (*p*-type) of *pnp* transistor has a reverse bias and receives hole charges that flow in the output circuit. Similarly, in Fig. (*ii*), the collector (*n*-type) of *npn* transistor has reverse bias and receives electrons.

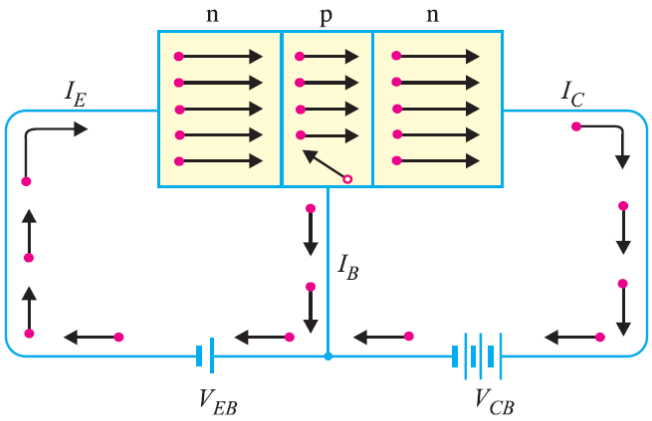
(iii) Base. The middle section which forms two pn-junctions between the emitter and collector is called the base. The base-emitter junction is forward biased, allowing low resistance for the emitter circuit. The base-collector junction is reverse biased and provides high resistance in the collector circuit.



Working Operation of NPN & PNP Transistor

Working of npn transistor. Fig. (iii) shows the npn transistor with forward bias to emitter base junction and reverse bias to collector-base junction. The forward bias causes the electrons in the n-type emitter to flow towards the base. This constitutes the emitter current I_E. As these electrons flow through the p-type base, they tend to combine with holes. As the base is lightly doped and very thin, therefore, only a few electrons (less than 5%) combine with holes to constitute base current I_B. The remainder (more than 95%) cross over into the collector region to constitute collector current I_C. In this way, almost the entire emitter current flows in the collector circuit. It is clear that, emitter current is the sum of collector and base currents i.e.

$$I_E = I_B + I_C$$



Basic connection of *npn* transistor

Fig. (iii) NPN Operation

Working of pnp transistor. Fig. (iv) shows the basic connection of a *pnp* transistor. The forward bias causes the holes in the *p*-type emitter to flow towards the base. This constitutes the emitter current I_E. As these holes cross into *n*-type base, they tend to combine with the electrons. As the base is lightly doped and very thin, therefore, only a few holes (less than 5%) combine with the electrons. The remainder (more than 95%) cross into the collector region to constitute collector current I_C. In this way, almost the entire emitter current flows in the collector circuit. It may be noted that current conduction within pnp transistor is by holes. However, in the external connecting wires, the current is still by electrons.

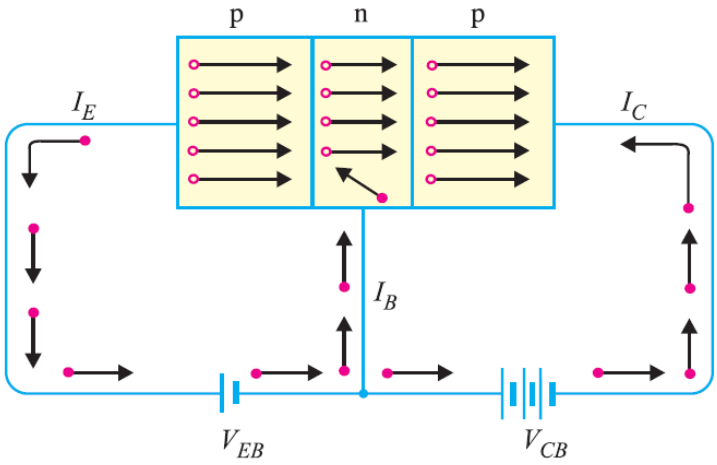
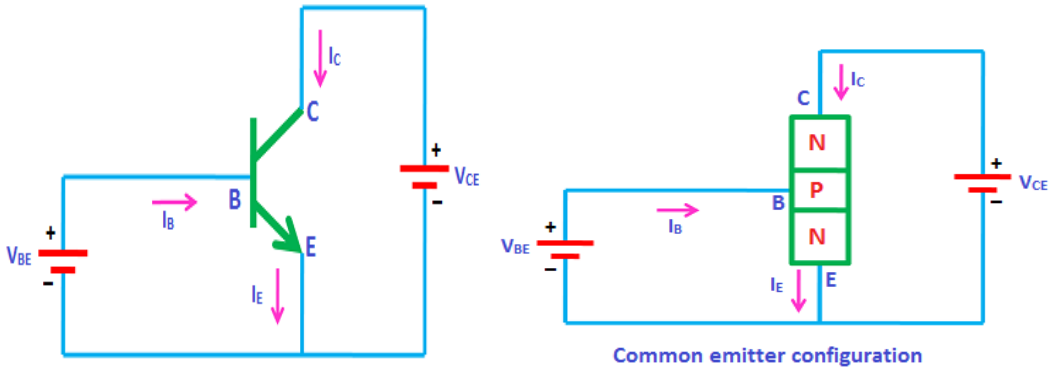


Fig. (iv) PNP Operation

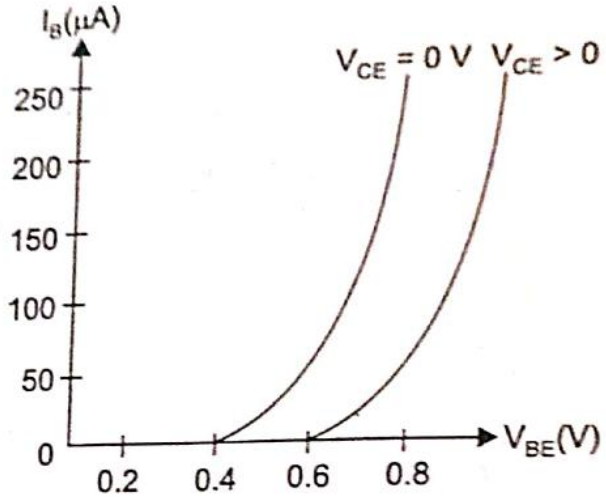
5. Explain **common emitter mode of transistor** & draw **input and output characteristics**.

In common emitter configuration, base is the input terminal, collector is the output terminal and emitter is common to both the terminals. This is also called as grounded emitter configuration. The circuit of common-emitter configuration is shown in figure .The

behavior of the transistor is analyzed with the help of the input characteristics and output characteristics.

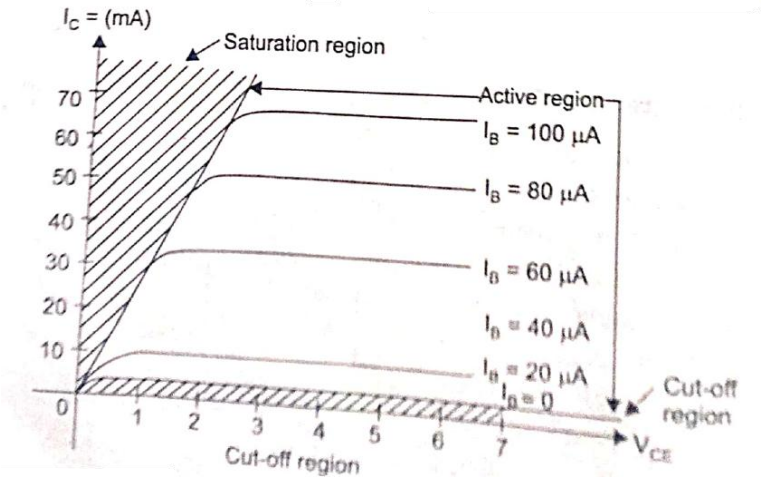


Input characteristics: The input characteristic curve defines the relation the base current I_B and the base to emitter voltage V_{BE} at a constant voltage V_{CE}. The input characteristics are obtained by increasing the base current in equal steps by increasing V_{BE} a fixed value of V_{CE}, zero voltage. When V_{CE} is zero, the emitter base junction is forward biased and the behaves as a forward biased diode. So, the input characteristics resemble the forward biased diode at V_{CE} =0. When V_{CE} is increased, the width of the depletion region at the reverse biased collector base junction will increase. This will reduce the effective width of the base and leads to decrease in the base current I_B. In order to provide the same amount of I_B,V_{BE} should be increased. Hence the curves appear to the right as V_{CE} increases.



Output Characteristics: The output characteristic curves defines the relation between the collector current I_C and collector to emitter voltage V_{CE} at a constant base current I_B. The output characteristics are obtained by the change in collector current I_C due to a change in V_{CE} at constant input current I_B. As we change the output voltage V_{CE} and the input voltage overline V_{BE}, this will affect the biasing of the both the junctions and in turn the operating

regions of the transistor. Hence, the output characteristics have three regions, namely, active region, saturation region and cut-off region.



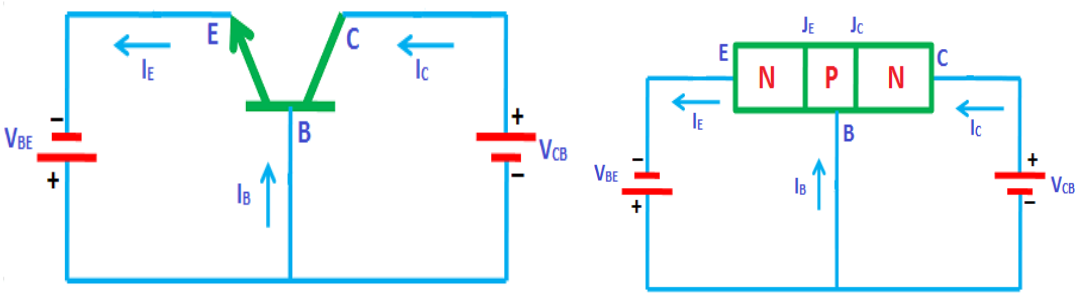
Base current amplification factor (β). In common emitter connection, input current is I_B and output current is I_C. The ratio of change in collector current (ΔIC) to the change in base current (ΔI_B) is known as base current amplification factor i.e.

$$\beta^* = \frac{\Delta I_C}{\Delta I_B}$$

6. Explain **common base mode of transistor** & draw **input, output characteristics**.

In a common-base transistor configuration, the emitter is the input terminal, the collector is the output terminal, and the base of the transistor is common to both terminals. This is also called the grounded base configuration. The circuit diagram of the common-base configuration is shown in figure.

In this circuit arrangement, input is applied between emitter and base and output is taken from collector and base. Base of the transistor is common to both input and output circuits and hence the name common base connection.

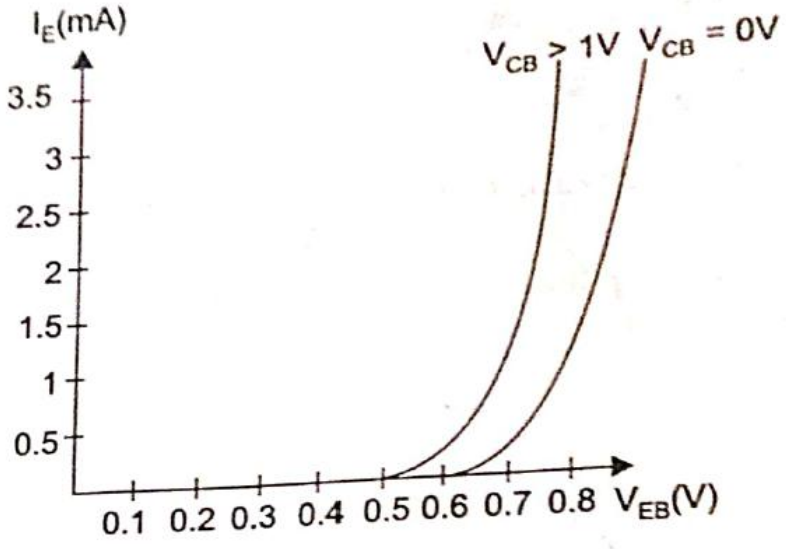


Input Characteristics:

The input characteristics curves define the relation between the input or emitter current I_E and emitter to base voltage V_{EB} at a constant output voltage V_{CB}. The input characteristics of the transistor are determined by keeping the output voltage collector –base voltage V_{CB} is kept constant at zero volt and the emitter current I_E is increased from zero in suitable steps by increasing V_{EB}. When V_{CB} is equal to zero, the emitter base junction is forward biased, and the junction behaves as a forward biased diode. The emitter current I_E increases rapidly with small increase in emitter base voltage V_{EB}. When V_{CB} is increased keeping V_{EB} constant, the width of the base region will decrease. Thus, small V_{EB} is enough to maintain the same I_E. This results an increase in I_E.

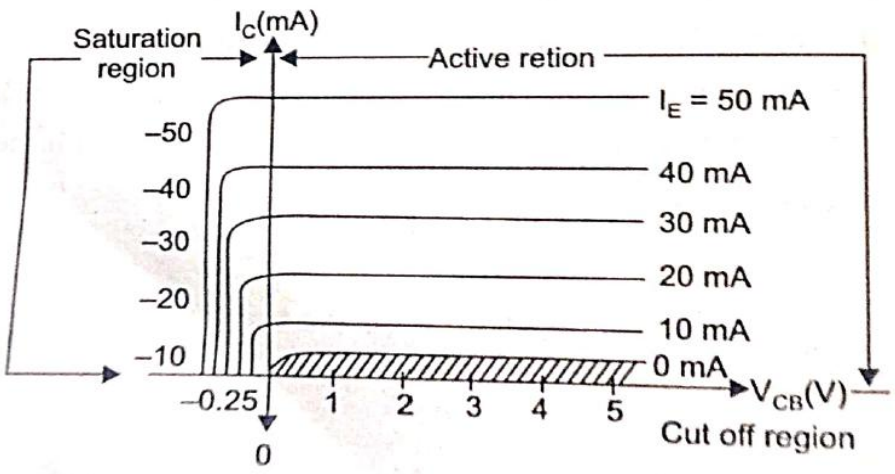
Input resistance. It is the ratio of change in emitter-base voltage (ΔV_{EB}) to the resulting change in emitter current (ΔI_E) at constant collector-base voltage (V_{CB}).

$$\text{Input resistance, } r_i = \frac{\Delta V_{BE}}{\Delta I_E} \quad \text{at constant } V_{CB}$$



Output characteristics.

It is the curve between collector current I_C and collector-base voltage V_{CB} at constant emitter current I_E. Generally, collector current is taken along y-axis and collector-base voltage along x-axis. Then V_{CB} is increased in uniform steps and the collector I_C is noted for each value of I_E. This is repeated for different values of I_E and is plotted as shown in figure (7.9). The characteristic curves shows that **at a constant value of I_E, I_C is independent of V_{CB}**. Moreover, I_C flows even when V_{CB} is zero.

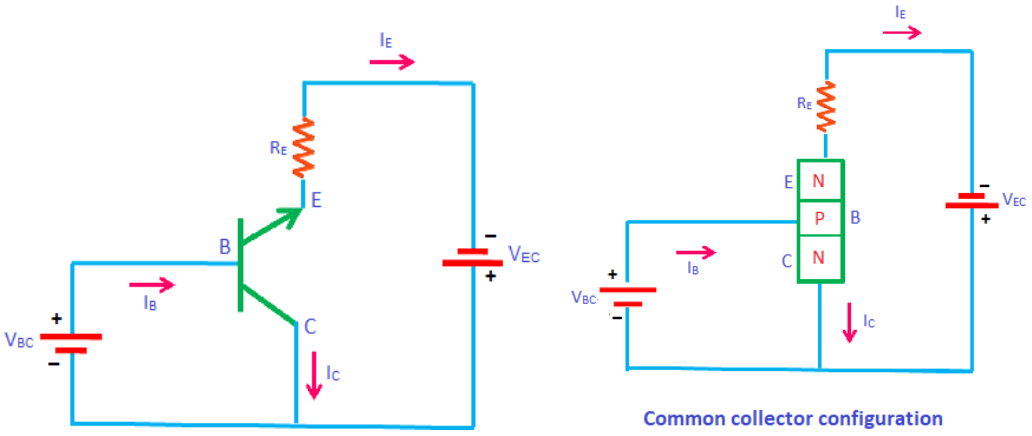


Output resistance. It is the ratio of change in collector-base voltage (ΔV_{CB}) to the resulting change in collector current (ΔI_C) at constant emitter current *i.e.*

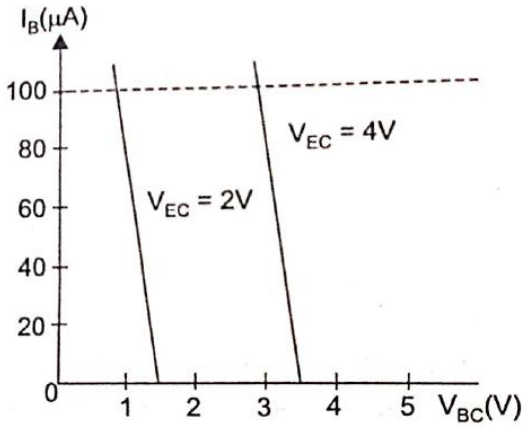
$$\text{Output resistance, } r_o = \frac{\Delta V_{CB}}{\Delta I_C} \text{ at constant } I_E$$

7. Explain **common-collector mode of transistor** & draw **input, output characteristics**.

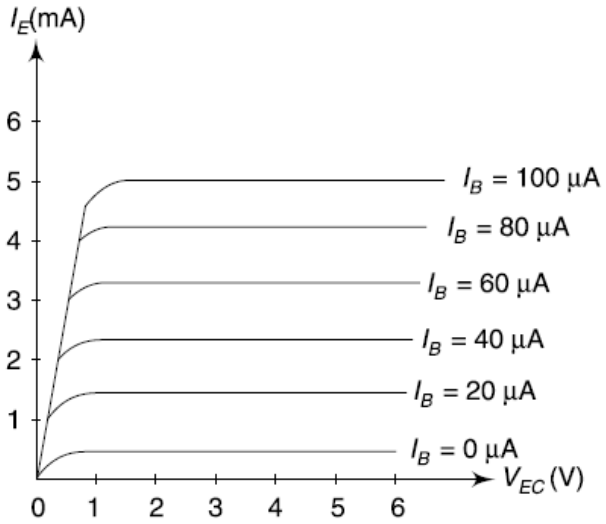
   In common collector configuration, the base is the input terminal, the emitter is the output terminal and collector is the common terminal. This is known as grounded collector configuration.



Input Characteristics: The input characteristic curves define the relation between the base current I_B and the base to collector voltage V_{BC} at a constant voltage V_{EC} .The input characteristics are obtained at a fixed voltage of V_{EC} by increasing the base collector voltage V_{BC} in uniform steps and the corresponding increase in I_B is noted. This process is repeated different values of V_{EC}.



Output Characteristics: The output characteristic curves define the relation between the emitter current I_E and emitter to collector voltage V_{EC} at a constant base current I_B. The family of characteristic curves is similar to the common emitter configuration as the output current, i.e. emitter current is almost equal to the collector current.



8. **Compare PN diode** with **zener diode** & write the **applications**.

Parameter	PN Junction Diode	Zener Diode
• Definition	• A two-terminal semiconductor device formed by joining a p-type semiconductor with an n-type semiconductor by alloying is known as PN junction diode.	• A Zener diode is a special type of two terminal semiconductor device which is optimized to work in breakdown region.
• Doping level of p-type & n-type semiconductors	• The level of doping of p-type and n-type semiconductors used for constructing PN junction diode is low.	• The highly doped p-type and n-type semiconductors are used for making Zener diode.

• Width of depletion layer	• As the PN junction diode has low doping level, thus the width of its depletion layer is more.	• The Zener diode is highly doped, hence its depletion region is narrow.
• Effect of reverse current	• The high reverse current can damage the PN junction diode.	• The Zener diode does not get damaged by the reverse current.
• Breakdown voltage	• In case PN junction diode, the breakdown voltage is high.	• The Zener diode has relatively lower breakdown voltage.
• Biasing condition	• The PN junction diode is designed to operate in forward bias only, i.e. current flows from anode to cathode.	• The Zener diode is mainly designed to operate in reverse biased conditions.
• Ohm's law	• The PN junction obeys Ohm's law.	• The Zener diode does not obey Ohm's law.
• Manufacturing materials	• The PN junction diode is generally made by silicon and germanium. Silicon is extensively used.	• The material used for making the Zener diode is selected according to the required voltage.
• Current flow	• In PN junction diode, the current flows only in one direction (or forward direction).	• In case of Zener diode, current can flow in both directions.
• Behave in forward & reverse bias	• A forward biased ideal PN junction diode acts a closed switch while in the reverse bias acts as an open switch, i.e. do not conduct.	• Zener diode acts as a normal PN junction diode in the forward bias condition, but it also conducts in the reverse bias condition.
• Application	• The PN junction diode extensively used in rectification, clamping, clipping, etc.	• The Zener diode is mainly used as a voltage regulator (or stabilizer) in the electronic devices and circuits.

All the Best